Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 270
Manufacturing Technology

Lecture 4: Fundamentals of Welding

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Lecture Objectives

- Understand the fundamentals of welding technology
- The weld joint
- Physics of welding
- Features of a fusion-welded joint

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Joining and Assembly

- Joining includes process such as welding, brazing, soldering, and adhesive bonding.
- Joining processes form a permanent joint between parts
- Assembly includes mechanical methods (usually) of fastening parts together.
- Some assembly processes allow for easy disassembly, while others do not.

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Welding

- It is a joining process in which two (or more) parts are coalesced at their contacting surfaces by application of heat and/or pressure
- Many welding processes are accomplished by heat alone, with no pressure applied
- Some welding processes involve the combination of heat and pressure while others involve pressure alone with no external heat.
- Some welding processes have a **filler** material added to facilitate coalescence.

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Welding In Ghana



Importance of welding

- It provides a permanent joint. Welded components become a single entity.
- It is usually the most economical way to join parts in terms of material usage and fabrication costs. Mechanical fastening usually requires additional hardware (e.g., screws) and geometric alterations of the assembled parts (e.g., holes)
- Welding is not restricted to a factory environment. It can be accomplished "in the field."

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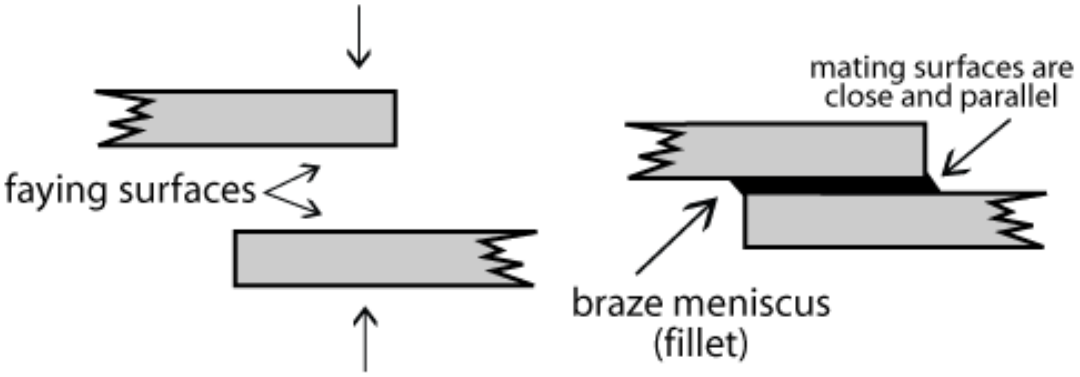
Problems associated with welding

- Most welding operations are performed manually and are expensive in terms of labour cost.
- Most welding processes utilize high energy and are inherently dangerous.
- Welded joints do not allow for convenient disassembly.
- Welded joints can have quality defects that are difficult to detect.

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Faying surfaces in welding

- These are the part surfaces in contact or close proximity that are being joined.
- Welding involves localized coalescence of the two metallic parts at their faying surfaces.
- **Welding is usually performed on parts made of the same metal. However, some welding operations can be used to join dissimilar metals.**



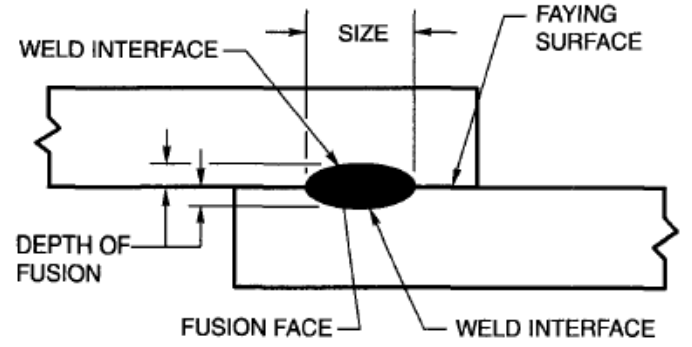


Figure R-6—Resistance Spot Weld

Types of welding processes

- Over 50 different types of welding processes have been catalogued by the American Welding Society (AWS)
- Welding processes can be divided into two major categories: **Fusion welding** and **Solid state welding.**

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Fusion welding

- This involves joining processes that melt the base metals
- In many fusion welding operations, a filler metal is added to the molten pool to facilitate the process and provide bulk and added strength to the welded joint.
- A fusion welding operation in which no filler metal is added is called an **autogenous** weld.

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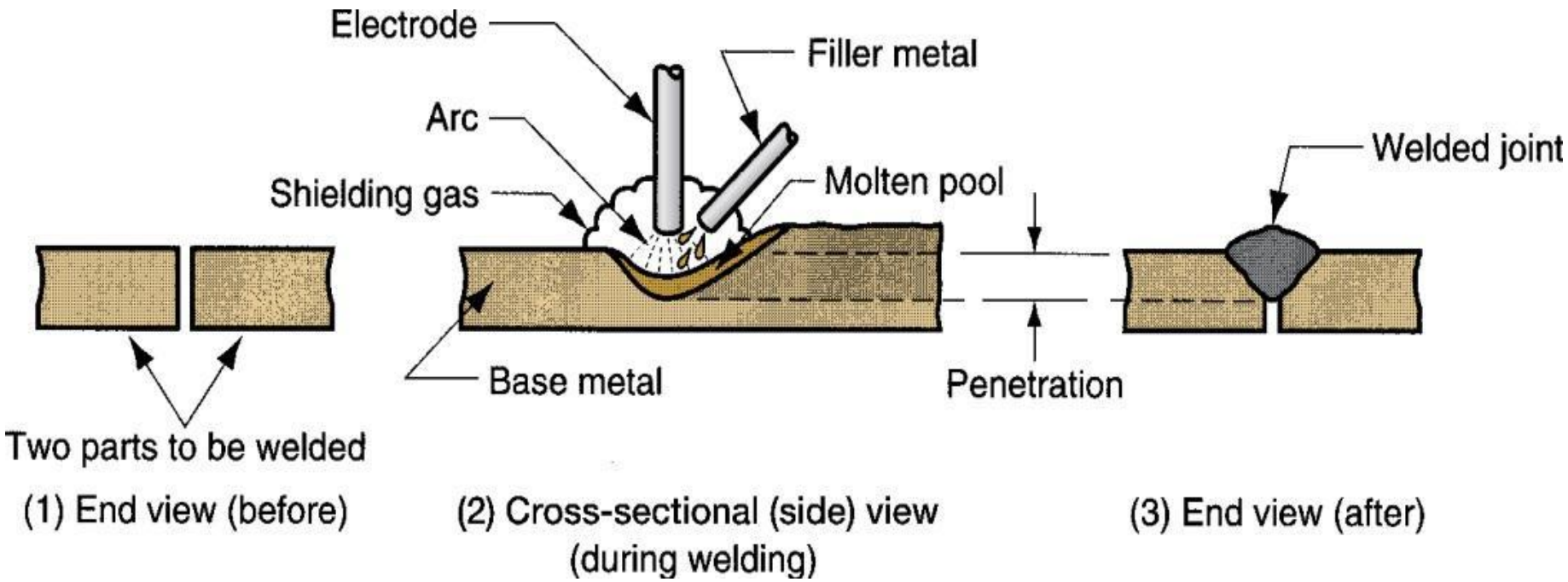
Sample fusion welding processes

- **Arc welding (AW)** – melting of the metals is accomplished by an electric arc.
- **Resistance welding (RW)** - melting is accomplished by heat from resistance to an electrical current between faying surfaces held together under pressure.
- **Oxyfuel gas welding (OFW)** - melting is accomplished by an oxyfuel gas such as acetylene.

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Arc welding

Basics of arc welding: (1) before the weld; (2) during the weld, the base metal is melted and filler metal is added to molten pool; and (3) the completed weldment.



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Solid state welding

- It is a joining process in which coalescence results from application of pressure alone or a combination of heat and pressure.
- If heat is used, temperature is below melting point of metals being welded.
- No filler metal is added in solid state welding.

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Solid state welding processes

- **Diffusion welding (DFW)** – coalescence is by solid state fusion between two surfaces held together under pressure at elevated temperature.
- **Friction welding (FRW)** - coalescence by heat of friction between two surfaces
- **Ultrasonic welding (USW)** - coalescence by ultrasonic oscillating motion in a direction parallel to contacting surfaces of two parts held together under pressure

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Principal applications of welding

- Construction - buildings and bridges
- Piping, pressure vessels, boilers, and storage tanks
- Shipbuilding
- Aircraft and aerospace
- Automotive
- Railroad

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Welder and Fitter

- The welder manually controls the path or placement of welding gun.
- Often assisted by second worker, called a ***fitter***, who arranges the parts prior to welding.
- Welding fixtures and positioners are used to assist in this function.

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Safety in Welding

- Welding is inherently dangerous to human workers.
- High temperatures of molten metals.
- In gas welding, fuels (e.g., acetylene) are a fire hazard.
- Many welding processes use electrical power, so electrical shock is a hazard.

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Hazards in Arc welding

- Ultraviolet radiation emitted in arc welding is injurious to human vision.
- Welder must wear special helmet with dark viewing window that filters out dangerous radiation. When using the helmet, the welder is blind except when arc is struck.
- Arc welding produces sparks, spatters of molten metal, smoke, and fumes. Ventilation is, therefore, needed to exhaust dangerous fumes from fluxes and molten metals.

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Automating the welding process

- Because of the hazards of manual welding, and to increase productivity and improve quality, various forms of mechanization and automation are used.
- **Machine welding** – It is a mechanized welding under supervision and control of human operator.
- **Automatic welding** – In this case, equipment performs welding without operator control.
- **Robotic welding** – This is an automatic welding process implemented by industrial robot.

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Automating the welding process



The weld joint

- Weld joint is the junction of the edges or surfaces of parts that have been joined by welding
- For weld joints, **the types of joints** and **types of welds used to join the pieces that form the joints** are important.

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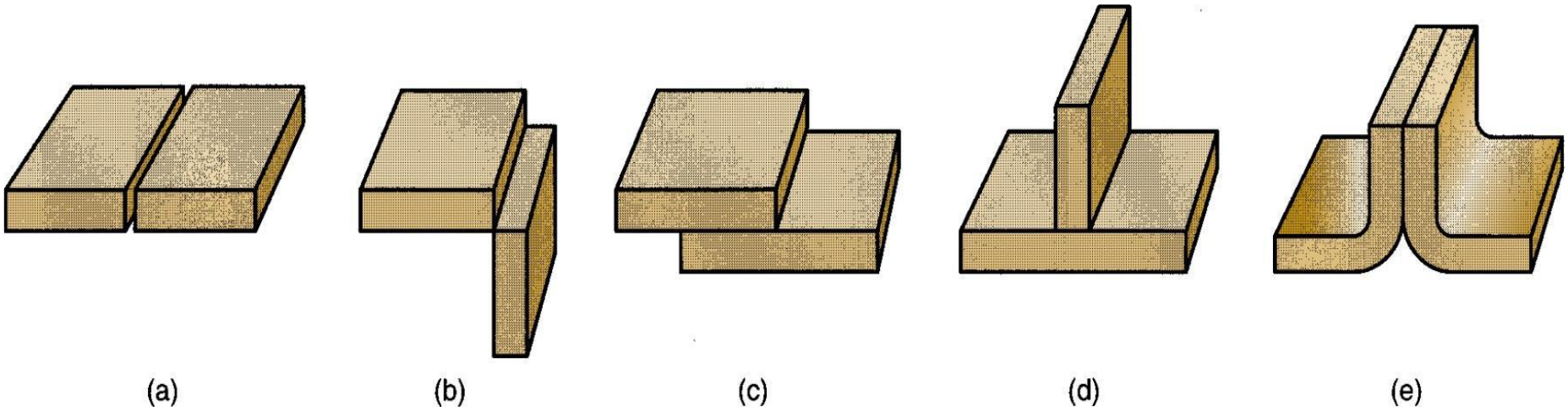
Weld Joints



Types of joints

- There are five (5) types of joints, namely

(a) Butt joint, (b) corner joint, (c) lap joint, (d) tee joint, and (e) edge joint



Types of joints

- **Butt joint**. In this joint type, the parts lie in the same plane and are joined at their edges.
- **Corner joint**. The parts in a corner joint form a right angle and are joined at the corner of the angle.
- **Lap joint**. This joint consists of two overlapping parts.
- **Tee joint**. In a tee joint, one part is perpendicular to the other in the approximate shape of the letter "T."
- **Edge joint**. The parts in an edge joint are parallel with at least one of their edges in common, and the joint is made at the common edge(s).

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Types of Weld

- Each of the preceding joints can be made by welding
- Other joining processes can also be used for some of the joint types
- There is a difference between joint type and the way it is welded - **the weld type.**

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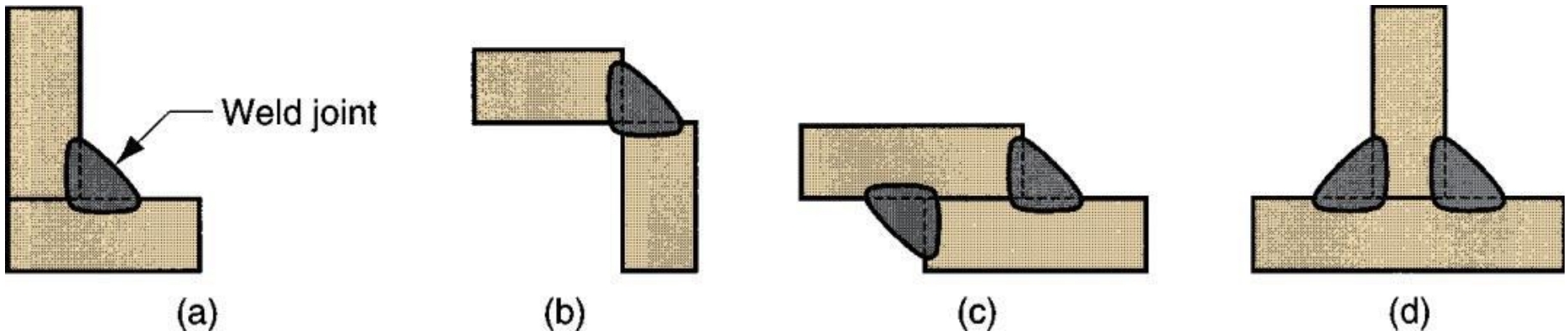
Fillet Weld

- Used to fill in the edges of plates created by corner, lap, and tee joints.
- Filler metal used to provide cross section in approximate shape of a right triangle.
- Most common weld type in arc and oxyfuel welding.
- Requires minimum edge preparation.

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Types of fillet welds

(a) Inside single fillet corner joint; (b) outside single fillet corner joint; (c) double fillet lap joint; (d) double fillet tee joint (dashed lines show the original part edges)



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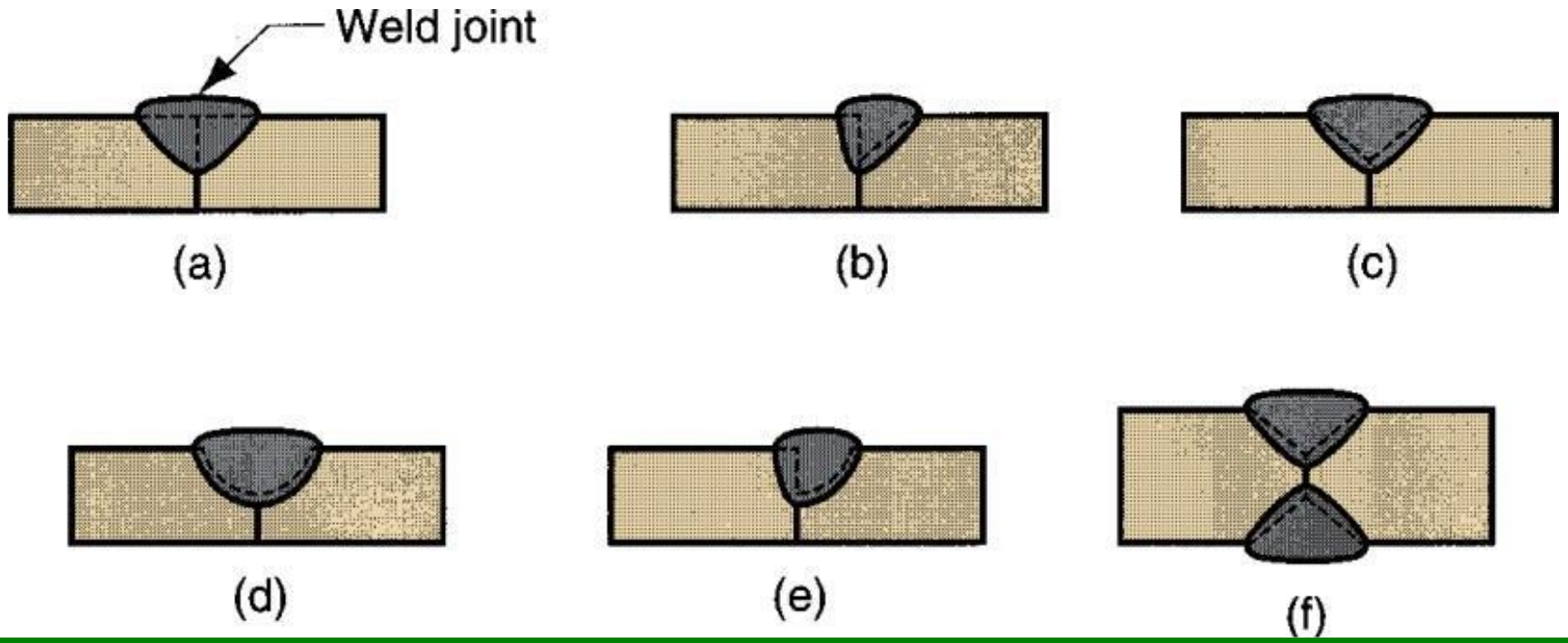
Groove welds

- Usually requires part edges to be shaped into a groove to facilitate weld penetration.
- Edge preparation increases cost of parts fabrication.
- Grooved shapes include square, bevel, V, U, and J, in single or double sides.
- Most closely associated with butt joints.

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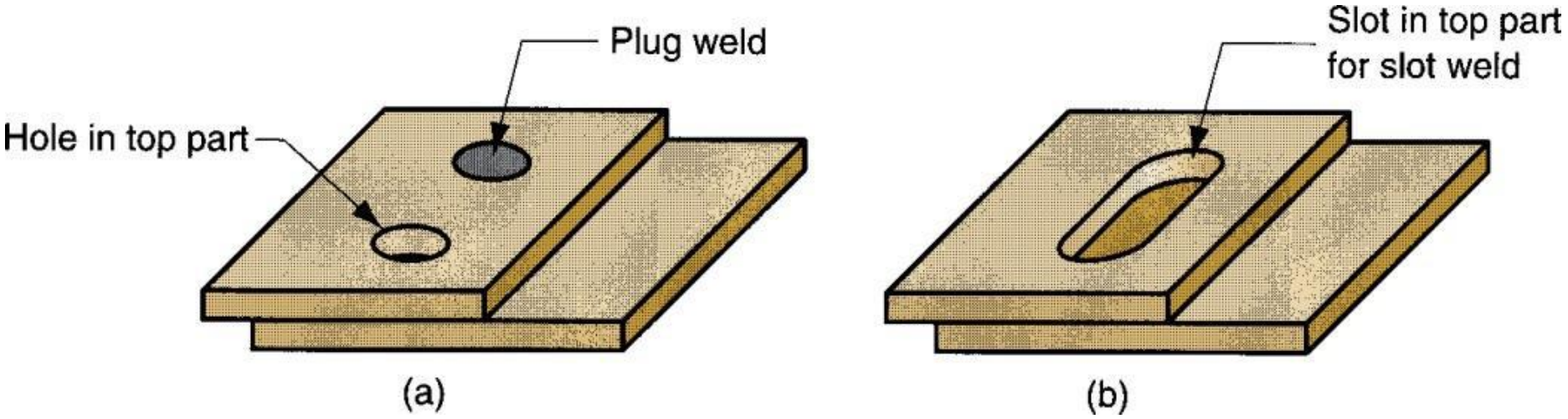
Groove welds

(a) Square groove weld, one side; (b) single bevel groove weld; (c) single V-groove weld; (d) single U-groove weld; (e) single J-groove weld; (f) double V-groove weld for thicker sections (dashed lines show original part edges).



Plug weld and slot weld

- Plug welds and slot welds are used for attaching flat plates, using one or more holes or slots in the top part and then filling with filler metal to fuse the two parts together.



(a) Plug weld and (b) slot weld

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Resistance welding

- Resistance welding is the method used to join two plain metal work pieces together by running an electrical current through them.
- The necessary welding heat is generated by the electrical resistance of the metals, by the contact resistance in between them and by the electrical current.
- Main forms include **spot welding and seam welding**.

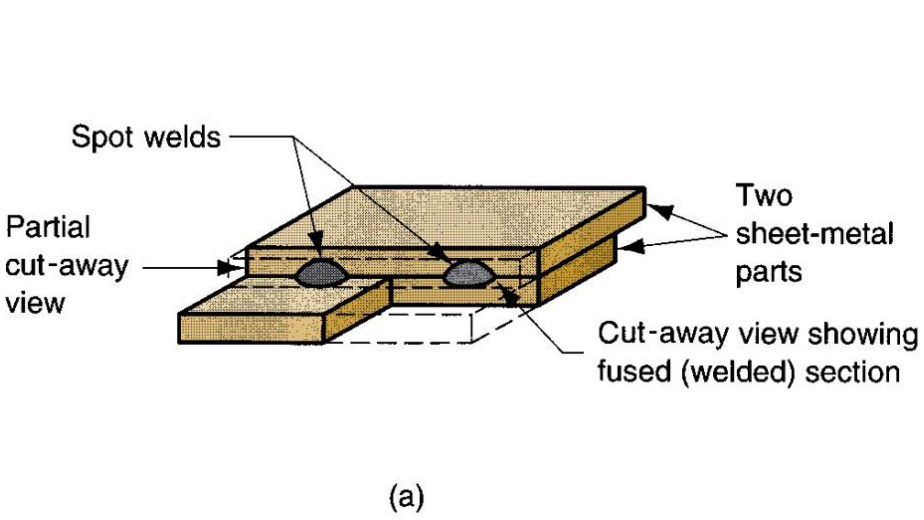
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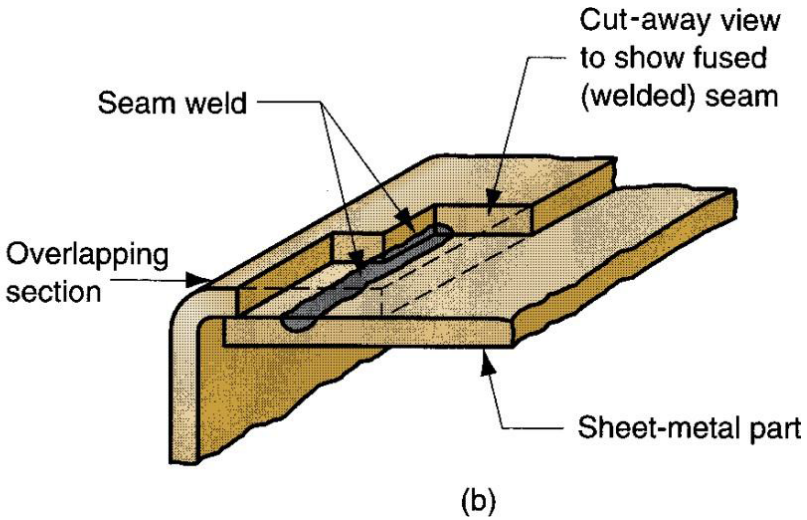
Spot welding and Seam welding



Spot Welding and Seam Welding

- Fused section between surfaces of two sheets or plates.
- Used for lap joints.
- Closely associated with resistance welding.





(a) spot weld and (b) seam weld

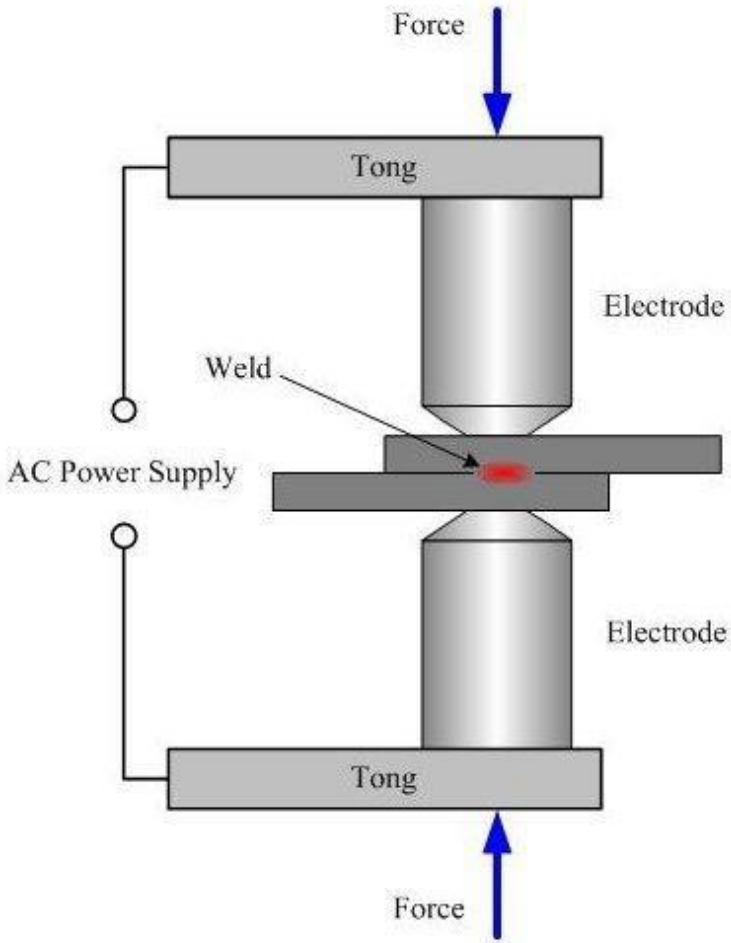
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Difference between seam welding and spot welding

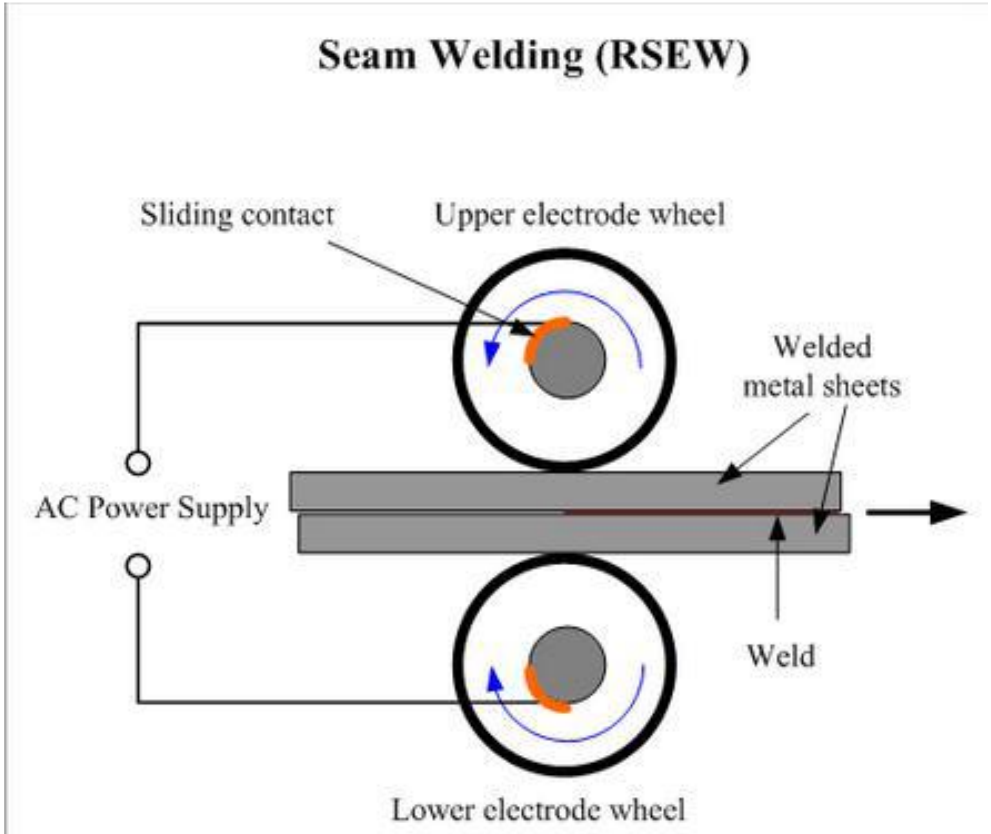
- Seam welding is similar to Spot welding.
- Difference lies in the use of rotating wheel in seam welding.
- Seam welding is faster than spot welding
- **Application:** Spot welding is mainly used for joining the body of automobiles while Seam welding is used for making fuel tank, transformers, paint containers etc.

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Difference between seam welding and spot welding



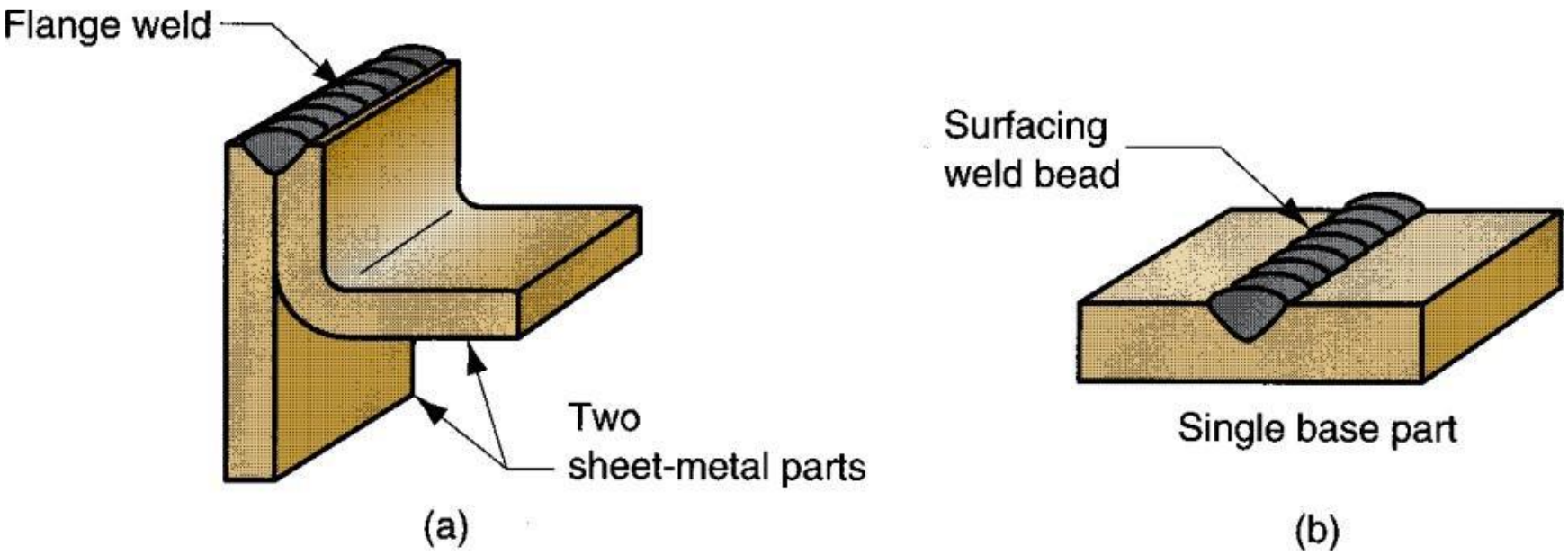
Spot welding



Seam welding

Flange Weld and Surfacing Weld

- Flange weld and surfacing weld are used not to join parts but to deposit filler metal onto surface of a base part.



(a) Flange weld and (b) Surfacing weld

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Physics of welding

- Fusion is most common means of achieving coalescence in welding
- To accomplish fusion, a source of high density heat energy must be supplied to the faying surfaces
- Resulting temperatures cause localized melting of base metals (and filler metal, if used)
- For metallurgical reasons, it is desirable to melt the metal with minimum energy but high heat densities

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Power Density

- It is defined as the power transferred to work per unit surface area, W/mm^2.
- If power density is too low, heat is conducted into work, so melting never occurs.
- If power density too high, localized temperatures vaporize metal in affected region.
- There is a practical range of values for power density within which welding can be performed.

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Comparison between welding processes

- Oxyfuel gas welding (OFW) develops large amounts of heat, but heat density is relatively low because heat is spread over a large area
- Oxyacetylene gas, the hottest OFW fuel, burns at a top temperature of around 3500 ℃ (6300 ℉)
- Arc welding produces high energy over a smaller area, resulting in local temperatures of 5500 ℃ to 6600 ℃ (10,000 ℉ to 12,000 ℉).

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Power Densities for Welding Processes

Welding process	W/mm^2	(Btu/sec-in^2)
Oxyfuel	10	(6)
Arc	50	(30)
Resistance	1,000	(600)
Laser beam	9,000	(5,000)
Electron beam	10,000	(6,000)

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Power Density

- Defined as power entering surface divided by corresponding surface area:

$$PD = \frac{P}{A}$$

Where PD = power density (W/mm^2)

P = Power entering surface (W)

A = surface area over which energy is entering (mm^2)

Unit Energy for Melting

- Quantity of heat required to melt a unit volume of metal
- Unit energy U_m is the sum of:
  - ✓ Heat to raise temperature of solid metal to melting point. (Depends on metal's volumetric specific heat)
  - ✓ Heat to transform metal from solid to liquid phase at melting point. (Depends on metal's heat of fusion)

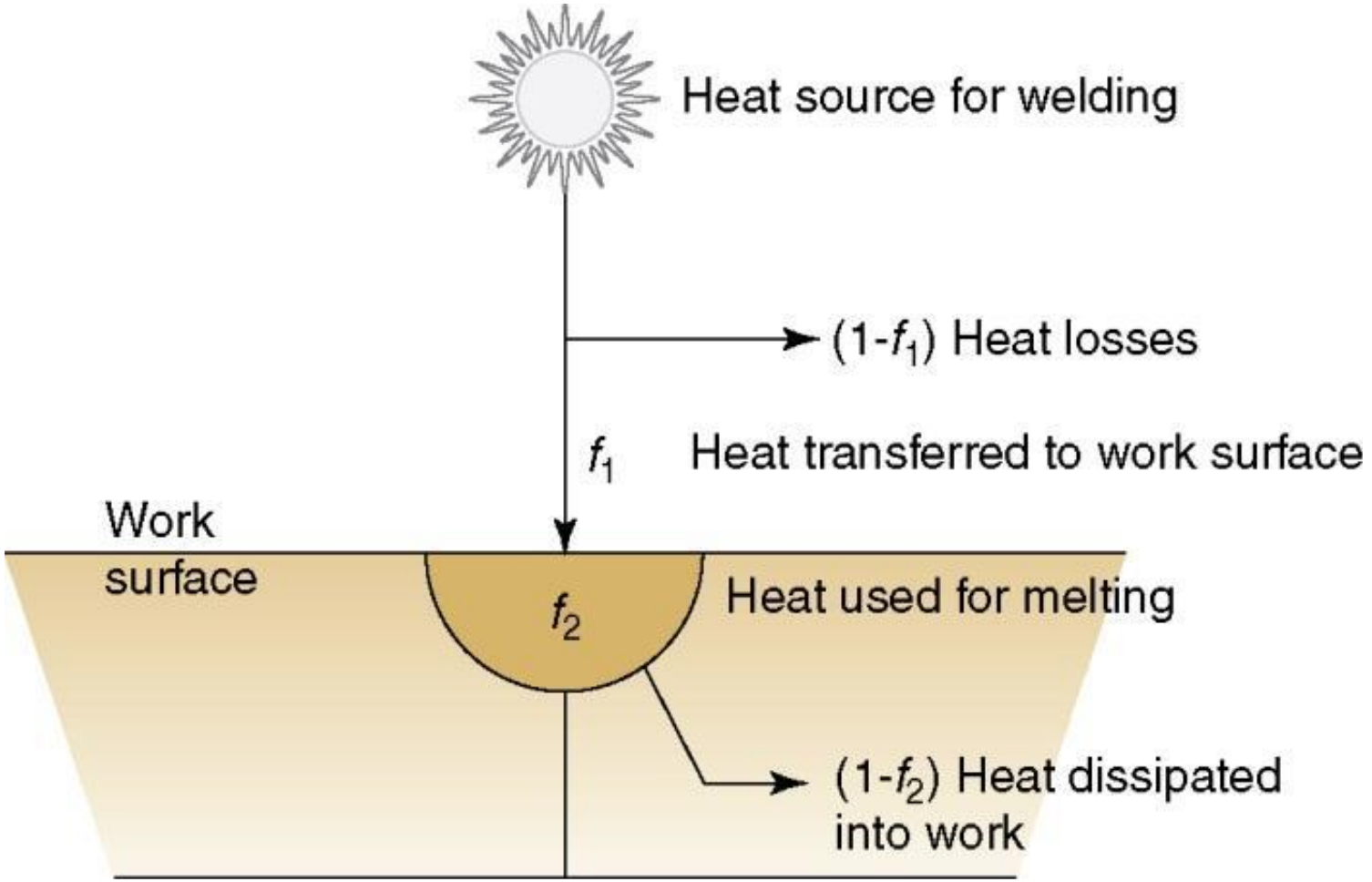
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Heat Transfer Mechanisms in Welding

- Not all of the input energy is used to melt the weld metal
  1. Heat transfer efficiency f_1 - actual heat received by workpiece divided by total heat generated at source.
  2. Melting efficiency f_2 - proportion of heat received at work surface used for melting.
- The rest is conducted into work metal.

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Heat Transfer Mechanisms in Welding



Heat Available for Welding

- The amount of heat available for welding is given by

$$H_w = f_1 f_2 H$$

where *Hw* is net heat available for welding

f_1 is heat transfer efficiency

f_2 is melting efficiency and

H is total heat generated by welding process

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Heat Transfer Efficiency f_1

- Proportion of heat received at work surface relative to total heat generated at source.
- Depends on welding process and capacity to convert power source (e.g., electrical energy) into usable heat at work surface
- Oxyfuel gas welding processes are relatively inefficient
- Arc welding processes are relatively efficient

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Melting Efficiency f_2

- It is the proportion of heat received at work surface used for melting; the rest is conducted into the work
- Depends on welding process but also thermal properties of metal, joint shape, and work thickness.
- Metals with high thermal conductivity, such as aluminium and copper, present a problem in welding because of the rapid dissipation of heat away from the heat contact area.

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Energy Balance Equation

- Net heat energy into welding operation equals heat energy required to melt the volume of metal welded

$$H_w = U_m V$$

where *Hw* is net heat energy delivered to operation (J)

Um is unit energy required to melt the metal (J/mm^3) and *V* is volume of metal melted (mm^3)

Example

A GTAW operation is performed on stainless steel, whose unit melting energy Um = 9.3 J/mm^3 . The conditions are: E = 25 volts, I = 125 amps, f_1 = 0.65, and f_2 = 0.70. If filler metal wire of 3.0 mm diameter is added to the operation, and the final weld bead is composed of equal volumes of 215 filler and base metal. If the travel speed in the operation v = 5 mm/sec, determine:

(a) cross-sectional area of the weld bead, and

(b) the feed rate (in mm/sec) at which the filler wire must be supplied.

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Solution

(a) Hw = $f_1 f_2$EI = $U_m A_w v$

$0.65(0.70)(25)(125) = 9.3(A_w)(5)$

$1421.9 = 46.5\ A_w$

Aw = $1421.9/46.5 = 30.6\ mm^2$

(b) $A_w v = 30.6(5) = 153\ mm^3/s$

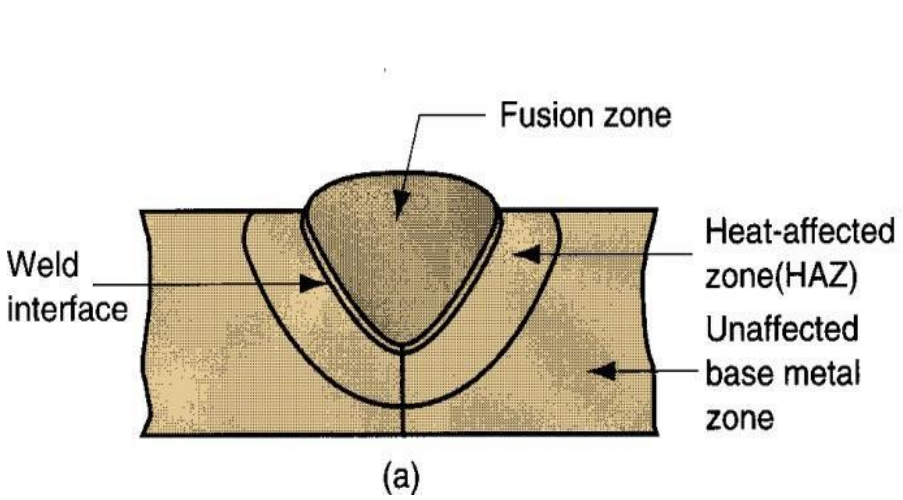
Filler wire A = $\pi D^2/_4 = \pi 3^2/_4 = 7.07\ mm^2$

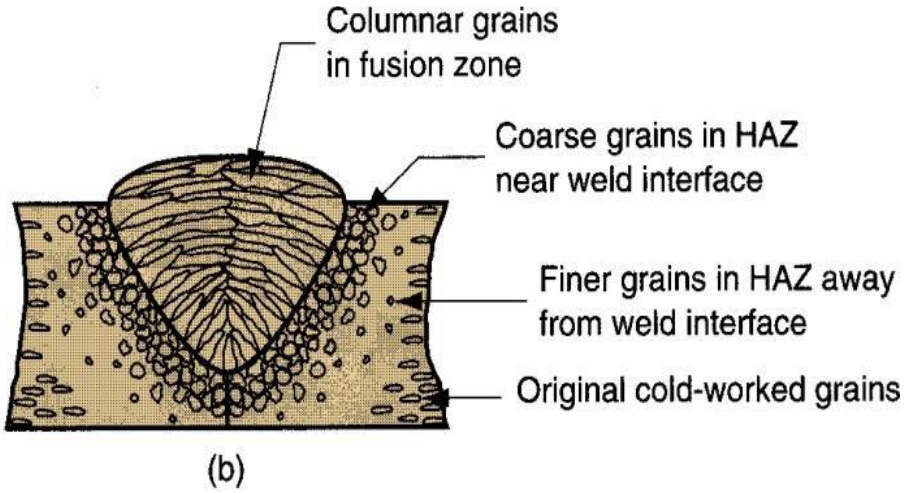
At 50% filler metal,

feed rate of filler wire = 153(0.50)/7.07 = 10.82 mm/s

Typical Fusion Welded Joint

- The figure below shows a cross section of a typical fusion welded joint





(a) principal zones in the joint

(b) typical grain structure

Features of Fusion Welded Joint

Typical fusion weld joint in which filler metal has been added consists of:

- Fusion zone
- Weld interface
- Heat affected zone (HAZ)
- Unaffected base metal zone

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Heat Affected Zone

- Metal has experienced temperatures below melting point, but high enough to cause microstructural changes in the solid metal.
- Chemical composition same as base metal, but this region has been heat treated so that its properties and structure have been altered.
- Effect on mechanical properties in HAZ is usually negative.
- It is here that welding failures often occur.

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Presentations

GROUPS	TOPIC
A	Arc welding
B	Resistance welding
C	Oxyfuel Gas fuel
D	Electron and Laser beam welding
E	Thermit welding
F	Solid state welding
G	Weld quality
H	Design considerations in welding

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